

APPLICATION OF DERIVATIVES

W.E-1. Find the equation of a tangent and normal at $x=0$ if they exist on the curve $y = x^{1/3}(1 - \cos x)$

[Hint: $\left. \frac{dy}{dx} \right|_{x=0} = 0$]

W.E-2. Prove that the portion of the tangent to the curve (tractrix)

$$\frac{x + \sqrt{a^2 - y^2}}{a} = \ln \frac{a + \sqrt{a^2 - y^2}}{y}$$

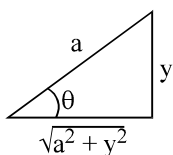
intercepted between the point of contact and the x-axis is constant.

[Ans. $l = a$]

[Hint: Substituting $y = a \sin \theta$ (1)]

$$\frac{x + a \cos \theta}{a} = \ln \frac{a + a \cos \theta}{a \sin \theta}$$

$$= \ln \frac{1 + \cos \theta}{\sin \theta} = \ln (\operatorname{cosec} \theta + \cot \theta)$$



$$\frac{x}{a} = \ln (\operatorname{cosec} \theta + \cot \theta) - \cos \theta$$

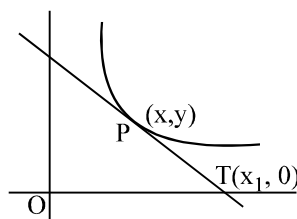
$$\frac{1}{a} \frac{dx}{d\theta} = - \frac{-\operatorname{cosec} \theta (\operatorname{cosec} \theta + \cot \theta)}{\operatorname{cosec} \theta + \cot \theta} +$$

$$\sin \theta = -\operatorname{cosec} \theta + \sin \theta = \sin \theta - \frac{1}{\sin \theta} = -\frac{\cos^2 \theta}{\sin \theta}$$

$$\therefore \frac{dx}{d\theta} = -\frac{a \cos^2 \theta}{\sin \theta} \quad \dots(1) \quad \text{also}$$

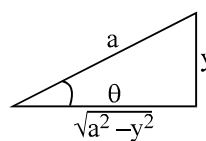
$$\frac{dy}{d\theta} = a \cos \theta \quad \dots(2)$$

$$\therefore \frac{dy}{dx} = \frac{a \cos \theta \sin \theta}{-a \cos^2 \theta} = -\tan \theta$$



equation of tangent
 $Y - y = -\tan \theta (X - x)$

$$Y - y = -\frac{y}{\sqrt{a^2 - y^2}} (X - x)$$

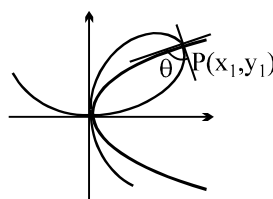


$$Y = 0 ; X = x + \sqrt{a^2 - y^2} = x_1 \text{ (say)}$$

$$PT^2 = a^2 - y^2 + y^2 \Rightarrow PT = a = \text{constant.}$$

W.E-3 :- Prove that the angle between the curves $y^2 = x$ and $x^3 + y^3 = 3xy$ at the point other than origin is $\tan^{-1}(16)^{1/3}$.

Sol.



$$y^2 = x ; x^3 + y^3 = 3xy$$

$$2y \frac{dy}{dx} = 1 \Rightarrow \left. \frac{dy}{dx} \right|_P = \frac{1}{2y_1}$$

again for the 2nd curve

$$\left. \frac{dy}{dx} \right|_P = \frac{y_1 - x_1^2}{y_1^2 - x_1}$$

solving $y^2 = x$ and $x^3 + y^3 = 3xy$;

$$y^6 + y^3 = 3y^3 \Rightarrow y^3 + 1 = 3 \Rightarrow y^3 = 2$$

$$\therefore y_1 = 2^{1/3} \text{ and } x_1 = 2^{2/3}$$

$$\text{now } m_1 = \frac{1}{\frac{1}{2 \cdot 2^{2/3}}} = \frac{1}{2^{1/3}};$$

$$m_2 = \frac{\frac{1}{2^{2/3}} - \frac{4}{2}}{\frac{2}{2^{2/3}} - 2^{2/3}} \rightarrow \infty$$

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{1 - \frac{m_1}{m_2}}{\frac{1}{m_2} + m_1} \right| =$$

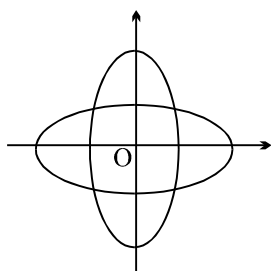
$$\left| \frac{1}{m_1} \right| = 2^{2/3} = 16^{1/3}$$

$$\therefore \theta = \tan^{-1}(16^{1/3})$$

W.E-4 :- Find the condition for the two concentric ellipses $a_1 x^2 + b_1 y^2 = 1$ and $a_2 x^2 + b_2 y^2 = 1$ to intersect orthogonally.

$$[\text{Ans. } \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{b_2} - \frac{1}{b_1}]$$

Sol.



$$a_1 x^2 + b_1 y^2 = 1 \quad \& \quad a_2 x^2 + b_2 y^2 = 1$$

$$2a_1 x + 2b_1 y > \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{a_1 x}{b_1 y}$$

$$\left. \frac{dy}{dx} \right|_{x_1, y_1}^I = -\frac{a_1 x_1}{b_1 y_1} \quad \dots(1)$$

$$\left. \frac{dy}{dx} \right|_{x_1, y_1}^{II} = -\frac{a_2 x_1}{b_2 y_1} \quad \dots(2)$$

If the two curves are orthogonal

$$\frac{a_1 x_1}{b_1 y_1} \cdot \frac{a_2 x_1}{b_2 y_1} = -1$$

$$\frac{a_1 a_2}{b_1 b_2} \frac{x_1^2}{y_1^2} = -1 \dots(3)$$

$$\text{now } a_1 x_1^2 + b_1 y_1^2 = 1 \quad \&$$

$$a_2 x_1^2 + b_2 y_1^2 = 1$$

(as x_1, y_1 satisfy the given equations)

subtracting

$$(a_1 - a_2) x_1^2 + (b_1 - b_2) y_1^2 = 0$$

$$\frac{x_1^2}{y_1^2} = -\frac{b_1 - b_2}{a_1 - a_2}$$

$$\text{putting in (3)} \quad \frac{a_1 a_2}{b_1 b_2} \frac{b_1 - b_2}{a_1 - a_2} = 1$$

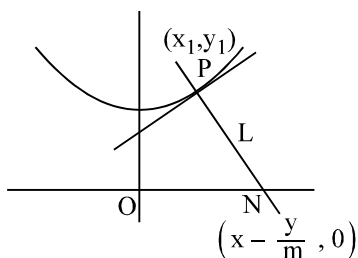
$$\frac{a_1 - a_2}{a_1 a_2} = \frac{b_1 - b_2}{b_1 b_2}$$

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{b_2} - \frac{1}{b_1}$$

W.E-5 :- Show that for the catenary $y = c \cosh\left(\frac{x}{c}\right)$

the length of the normal at any point is $\frac{y^2}{c}$.

$$\text{Sol. } y = c \cosh \frac{x}{c} = c \left(\frac{e^{\frac{x}{c}} + e^{-\frac{x}{c}}}{2} \right)$$



$$\frac{dy}{dx} = \frac{e^{\frac{x}{c}} - e^{-\frac{x}{c}}}{2} = m_T$$

$$\therefore m_N = \frac{-2}{e^{\frac{x}{c}} - e^{-\frac{x}{c}}} = m \text{ say}$$

equation of normal $Y - y = m(X - x)$

Put $Y = 0$ $X = x - \frac{y}{m}$

$$L^2 = \frac{y^2}{m^2} + y^2 = y^2 \left[1 + \frac{1}{m^2} \right]$$

$$= y^2 \left[1 + \frac{(e^{x/c} - e^{-x/c})^2}{4} \right]$$

$$= y^2 \left[\frac{4 + (e^{x/c} - e^{-x/c})^2}{4} \right]$$

$$= y^2 \left[\left(\frac{e^{x/c} + e^{-x/c}}{2} \right)^2 \right]$$

$$\Rightarrow L = y \left(\frac{e^{x/c} + e^{-x/c}}{2} \right) = y \cdot \frac{y}{c} = \frac{y^2}{c}$$

W.E-6 :- Show that for the curve $y^2 = (x + a)^3$ the square of the subtangent varies as the subnormal.

Hint: TPT $\left(\frac{y}{(dy/dx)} \right)^2 = k \cdot y \left(\frac{dy}{dx} \right)$ or

$$y = k \left(\frac{dy}{dx} \right)^3$$

now $2by \frac{dy}{dx} = 3(x + a)^2 \Rightarrow \frac{dy}{dx} = \frac{3(x + a)^2}{2by}$

hence $\frac{y}{(dy/dx)^3} = \frac{y 8b^3 y^3}{2y(x + a)^6}$

$$= \frac{8b^3}{2y} \left(\frac{y^2}{(x + a)^3} \right)^2 = \frac{8b^3}{27} \cdot \frac{1}{b^2}$$

$$= \frac{8b}{27}$$

W.E-7:- In an acute triangle ABC if sides a, b be constants and the base angles A and B vary, show that

$$\frac{dA}{\sqrt{a^2 - b^2 \sin^2 A}} = \frac{dB}{\sqrt{b^2 - a^2 \sin^2 B}}$$

Sol.

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

or $b \sin A = a \sin B$

$$b \cos A dA = a \cos B dB$$

$$\frac{dA}{a \cos B} = \frac{dB}{b \cos A} \Rightarrow$$

$$\frac{dA}{a \sqrt{1 - \sin^2 B}} = \frac{dB}{b \sqrt{1 - \sin^2 A}}$$

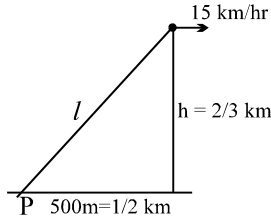
$$\frac{dA}{a \sqrt{1 - \frac{b^2 \sin^2 A}{a^2}}} = \frac{dB}{b \sqrt{1 - \frac{a^2 \sin^2 B}{b^2}}}$$

$$\frac{dA}{\sqrt{a^2 - b^2 \sin^2 A}} = \frac{dB}{\sqrt{b^2 - a^2 \sin^2 B}}$$

W.E-8 :- An aeroplane is flying horizontally at a height

of $\frac{2}{3}$ km with a velocity of 15 km/hr. Find the rate at which it is receding from a fixed point on the ground which it passed over 2 minutes ago. [Ans. 9 km/hr]

Sol.



$$l^2 = h^2 + x^2$$

$$2l \frac{dl}{dt} = 0 + 2x \frac{dx}{dt} \Rightarrow \frac{dl}{dt} = \frac{x}{l} \cdot \frac{dx}{dt}$$

where $x = \frac{1}{2}$ km, $h = \frac{2}{3}$ km then

$$l = \frac{1}{4} + \frac{4}{9} = \frac{5}{6} \text{ km}$$

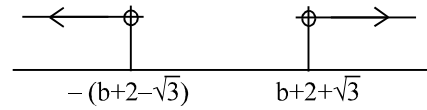
$$\therefore \frac{dl}{dt} = \frac{1}{2} \cdot \frac{6}{5} \cdot 15 = 9 \text{ km/hr}$$

W.E-9 :- Find all possible 'b' for which $f(x) = \sin 2x - 8(b+2)\cos x - (4b^2 + 16b + 6)x$ is monotonic decreasing for $\forall x \in \mathbb{R}$ and has no critical point.

$$[\text{Ans. } (-\infty, -(3+\sqrt{3})) \cup (\sqrt{3}-1, \infty)]$$

Sol. $f'(x) = 2 \cos 2x + 8(b+2) \sin x - (4b^2 + 16b + 6)$
 $= 2(1 - 2 \sin^2 x) + 8(b+2) \sin x - (4b^2 + 16b + 6)$
 for monotonic decreasing and no critical points
 $f'(x) < 0 \quad \forall x \in \mathbb{R}$
 hence $\sin^2 x - 2(b+2) \sin x + b^2 + 4b + 1 > 0$
 $[\sin x - (b+2)]^2 - (b+2)^2 + 4b + b^2 + 1 > 0$
 $[\sin x - (b+2)]^2 - (\sqrt{3})^2 > 0$
 $[\sin x - b - 2 - \sqrt{3}][\sin x - b - 2 + \sqrt{3}] > 0$
 $[\sin x - (b+2+\sqrt{3})][\sin x - (b+2-\sqrt{3})] > 0$

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Hence $\sin x > b + (2 + \sqrt{3})$

or $b + (2 + \sqrt{3}) < \sin x \quad \forall x \in \mathbb{R}$

$$\therefore b + (2 + \sqrt{3}) < -1$$

$$b < -(3 + \sqrt{3}) \Rightarrow b \in (-\infty, -(3 + \sqrt{3}))$$

|| ly $\sin x < (b + 2 - \sqrt{3})$

or $(b + 2 - \sqrt{3}) > \sin x \quad \forall x \in \mathbb{R}$

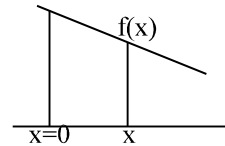
$$b + 2 - \sqrt{3} > 1$$

$$b > -1 + \sqrt{3}$$

$$b > \sqrt{3} - 1 \Rightarrow b \in ((\sqrt{3} - 1), \infty)$$

W.E- 10 :- Let f, g be differentiable on $\mathbb{R}$ and suppose that $f(0) = g(0)$ and $f'(x) \leq g'(x)$ for all $x \geq 0$. Show that $f(x) \leq g(x)$ for all $x \geq 0$.

Sol.



Given $f'(x) \leq g'(x)$ and $f(0) = g(0)$

TPT $f(x) \leq g(x) \quad \forall x \geq 0$

consider a function

$$F(x) = f(x) - g(x)$$

$$F'(x) = f'(x) - g'(x) \leq 0$$

hence $F(x)$ is a decreasing function for $x \geq 0$

$$\therefore F(x) \leq F(0) \quad \forall x \geq 0$$

but $F(0) = f(0) - g(0) = 0$ (given)

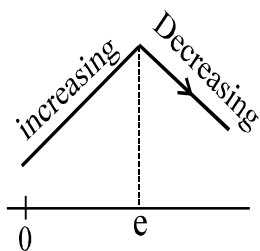
$$\therefore F(x) \leq 0 \quad \forall x \geq 0$$

$$\therefore f(x) - g(x) \leq 0 \quad \forall x \geq 0$$

$$\therefore f(x) \leq g(x)$$

W.E- 11 :- Use the function $f(x) = \frac{1}{x^x}$ ($x > 0$) to ascertain whether π^e or e^π is greater.

Sol.



$$f(x) = x^{\frac{1}{x}}; f(x) = e^{\frac{\ln x}{x}}$$

$$\Rightarrow f'(x) = x^{\frac{1}{x}} \left[\frac{1 - \ln x}{x^2} \right]$$

for $0 < x < e$, $f'(x) > 0$;
 $x > e$, $f'(x) < 0$

$f(x)$ attains the maximum at $x = e$

$$\therefore f(e) > f(x) \text{ for all } x > 0; e^{\frac{1}{e}} > e^{\frac{1}{x}}$$

$$\therefore e^{\frac{1}{e}} > \pi^{\frac{1}{\pi}} \quad e^{\pi} > \pi^e$$

W.E- 12 :- If $f(x) = |px - q| + r|x|$, $x \in (-\infty, \infty)$, $p, q, r > 0$, assumes its minimum value only at one point then find the relation between p, q, r .

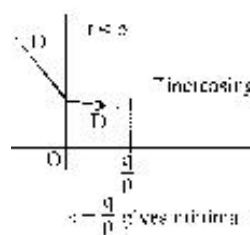
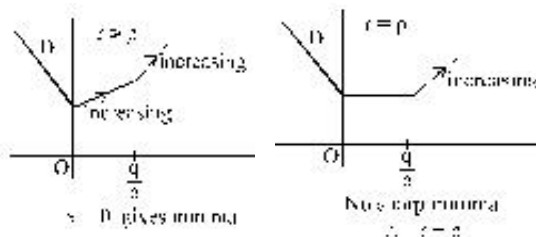
Sol. Let $x \geq \frac{q}{p}$; $f(x) = px - q + rx$;

$$f'(x) = p + r > 0 \Rightarrow f(x) \text{ increasing for } x \geq \frac{q}{p}$$

$$0 \leq x < \frac{q}{p} \quad f(x) = q - px + rx ;$$

$$f'(x) = r - p \Rightarrow \begin{cases} f(x) \text{ is } \uparrow \text{ if } r > p \\ \text{const} & r = p \\ f(x) \text{ is } \downarrow \text{ if } r < p \end{cases}$$

$$x < 0, f(x) = q - px - rx ; f'(x) = -(p+r) \\ \Rightarrow f(x) \text{ is decreasing}$$

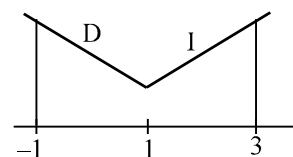


W.E- 13 :- Find the image of interval $[-1, 3]$ under the mapping specified by the function

$$f(x) = 4x^3 - 12x$$

Ans. $[-8, 72]$

Sol.



$$f'(x) = 0 \Rightarrow x = \pm 1$$

Find $f(-1)$, $f(1)$ and $f(3)$ and interpret.

W.E- 14 :- Let $f(x) = x^3 - x^2 + x + 1$ and

$$g(x) = \begin{cases} \max(f(t) : 0 \leq t \leq x), & 0 \leq x \leq 1 \\ 3 - x & \text{for } 1 < x \leq 2 \end{cases}$$

Check the continuity and differentiability of $g(x)$ in $(0, 2)$.

Sol. $f'(t)$ is always ($\uparrow$) increasing hence

$$g(x) = \begin{cases} x^3 - x^2 + x + 1 & \text{for } -1 \leq x \leq 1 \\ 3 - x & \text{for } 1 < x \leq 2 \end{cases}$$

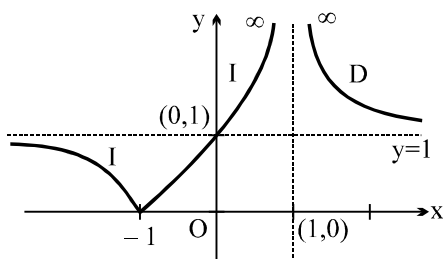
W.E- 15 :- Find the greatest and least value of the

$$\text{function } f(x) = \left| \frac{1+x}{1-x} \right| \text{ in } [-2, 0] \text{ and also}$$

sketch the graph of $f(x)$ and find its range.

Sol.
$$f(x) = \begin{cases} \frac{1+x}{1-x} & \text{if } -1 \leq x < 1 \\ \frac{x+1}{x-1} & \text{if } x > 1 \text{ or } x < -1 \end{cases}$$

$$f'(x) = \begin{cases} \frac{2}{(1-x)^2} & \uparrow \text{ in } (-1, 1) \\ \frac{-2}{(x-1)^2} & \downarrow \text{ in } x > 1 \text{ or } x < -1 \end{cases}$$



Range in $[0, \infty)$. Maximum value in $[-2, 0]$ occurs at $x = 0$ and $f(0) = 1$;

Minimum value in $[-2, 0]$ occurs at $x = -1$ and $f(-1) = 0$

W.E- 16 :- $\frac{\tan x}{x} > \frac{x}{\sin x}$ for $0 < x < \frac{\pi}{2}$.

Sol. We have to show that

$$\frac{\tan x \sin x - x^2}{x \sin x} > 0 \text{ for } 0 < x < \pi/2$$

Since $x \sin x > 0$ for $0 < x < \pi/2$, it is enough to show that

$$\tan x \cdot \sin x - x^2 > 0, \quad 0 < x < \pi/2$$

$$\text{Let } f(x) = \tan x \sin x - x^2 \text{ for } 0 < x < \pi/2$$

$$f'(x) = \sin x \sec^2 x + \tan x \cos x - 2x$$

$$= \sin x \sec^2 x + \sin x - 2x$$

$$f''(x) = \cos x \sec^2 x + \sin x \cdot 2 \sec^2 x \tan x + \cos x - 2$$

$$= \sec x + \cos x - 2 + 2 \sin x \tan x \sec^2 x$$

$$= (\sqrt{\sec x} - \sqrt{\cos x})^2 + 2 \sin x \tan x \sec^2 x$$

$$> 0 \text{ for } 0 < x < \pi/2$$

$\therefore f'$ is strictly increasing in $[0, \pi/2)$.

$$\text{Also } f'(0) = 0$$

$$\Rightarrow f'(x) > 0 \text{ for } 0 < x < \pi/2$$

$\Rightarrow f$ is strictly increasing in $[0, \pi/2)$.

$$\text{Also } f(0) = 0 \Rightarrow f(x) > 0 \text{ for } 0 < x < \pi/2.$$

$$\Rightarrow \tan x \sin x - x^2 > 0 \text{ for } 0 < x < \pi/2.$$

$$\therefore \frac{\tan x \sin x - x^2}{x \sin x} > 0, \quad 0 < x < \pi/2.$$

$$\Rightarrow \frac{\tan x}{x} > \frac{x}{\sin x} \text{ for } 0 < x < \frac{\pi}{2}.$$

W.E- 17 :- For $0 \leq p \leq 1$ and for any $a > 0, b > 0$.
 $(a + b)^p \leq a^p + b^p$.

Sol. $a^p \left(1 + \frac{b}{a}\right)^p \leq a^p + b^p;$

$$\left(1 + \frac{b}{a}\right)^p \leq 1 + \left(\frac{b}{a}\right)^p$$

$$[1 + x]^p \leq 1 + x^p \quad \text{where } x = b/a$$

W.E- 18 :- Show that between any two roots of the equation $e^x \cos x = 1$ there exists at least one root of $e^x \sin x - 1 = 0$.

Hint: Consider $f(x) = \cos x - e^{-x}$ and apply Rolle's theorem in it.

W.E- 19:- Let $f(x)$ and $g(x)$ be differentiable functions such that $f'(x)g(x) \neq f(x)g'(x)$ for any real x . Show that between any two real solutions of $f(x) = 0$, there is at least one real solution of $g(x) = 0$.

Sol. Let a, b be the solutions of $f(x) = 0$ suppose $g(x)$ is not equal to zero for any x belonging to $[a, b]$

$$\text{now consider } h(x) = f(x)/g(x)$$

since $g(x)$ not equal to zero

$h(x)$ is differentiable and continuous in (a, b)

$h(a) = h(b) = 0$ (as $f(a) = 0$ and $f(b) = 0$ but $g(a)$ or $g(b) \neq 0$)

applying rolles theorem

$h'(c) = 0$ for some c belonging to (a, b)

$$f(x)g'(x) = f'(x)g(x)$$

this gives the contradiction ...

W.E- 20 :- Let $P(x)$ be a polynomial with real coefficients. Let $a, b \in \mathbb{R}, a < b$, be two

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consecutive roots of $P(x)$. Show that there exists 'c' such that $a \leq c \leq b$ and

$$P'(c) + 100P(c) = 0.$$

Sol. Consider $f(x) = e^{100x} \cdot P(x)$.

$$\text{Now } f(a) = f(b) = 0$$

$$\{\text{as } P(a) = P(b) = 0\}$$

Also as $P(x)$ is polynomial $\Rightarrow f(x)$

is continuous and differentiable in $[a, b]$

$\Rightarrow$ Rolle's theorem can be applied

$\Rightarrow \exists c \in (a, b)$ such that $f'(c) = 0$

$$\text{now } f'(x) = e^{100x} (P'(x) + 100 \cdot P(x))$$

$$\Rightarrow e^{100c} (P'(c) + 100 \cdot P(c)) = 0,$$

from (1)

$$\Rightarrow P'(c) + 100 \cdot P(c) = 0 \text{ (as } [e^{100c} \neq 0])$$

W.E- 21 :- Using LMVT

prove that $|\cos a - \cos b| \leq |a - b|$

Hint: Consider $f(x) = \cos x$ in $[a, b]$

$$\left| \frac{\cos b - \cos a}{b - a} \right| = |-\sin C| \leq 1$$

$$\Rightarrow |\cos b - \cos a| \leq |b - a|$$

$$|\cos a - \cos b| \leq |a - b|$$

W.E- 22 :- Let a, b, c be three real number such that $a < b < c$, $f(x)$ is continuous in $[a, c]$ and differentiable in (a, c) . Also $f'(x)$ is strictly increasing in (a, c) . Prove that

$$(b - c)f(a) + (c - a)f(b) + (a - b)f(c) < 0$$

[Sol. By LMVT in $[a, b]$ for f

$$\frac{f(b) - f(a)}{b - a} = f'(u) \text{ where } a < u < b$$

and applying LMVT in $[b, c]$ and

$$\frac{f(c) - f(b)}{c - b} = f'(v) \text{ where } b < v < c$$

Since $f'(x)$ is strictly increasing

$$\therefore f'(u) < f'(v) ;$$

$$\therefore \frac{f(b) - f(a)}{b - a} < \frac{f(c) - f(b)}{c - b}$$

$$f(b)(c - b + b - a) - f(a)(c - b) - f(c)(b - a) < 0$$

$$\therefore (b - c)f(a) + (c - a)f(b) + (a - b)f(c) < 0$$

W.E- 23 :- If the function $f: [0, 4] \rightarrow \mathbb{R}$ is differentiable then show that for $a, b \in (0, 4)$

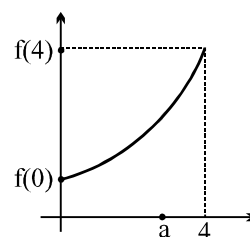
$$f^2(4) - f^2(0) = 8f'(a)f(b)$$

Sol. Since f is differentiable on $[0, 4]$ so it is continuous also. Hence by LMVT,

$\exists$ some $a \in (0, 4)$ such that

$$f'(a) = \frac{f(4) - f(0)}{4 - 0} \quad \dots(1)$$

again, by intermediate value theorem for continuous functions, $\exists b \in (0, 4)$ such that



$$f(b) = \frac{f(0) + f(4)}{2} \quad \dots(2)$$

from (1) and (2)

$$f'(a) \cdot f(b) = \frac{f^2(4) - f^2(0)}{8}$$

W.E- 24 :- If f and F are continuous in $[a, b]$ and derivable in (a, b) with $F'(x) \neq 0$

$\forall x \in (a, b)$. Prove that $\exists c \in (a, b)$ such

$$\text{that } \frac{f'(c)}{F'(c)} = \frac{f(b) - f(a)}{F(b) - F(a)}.$$

Sol.

$$\text{Let } K_1 = f(b) - f(a) \text{ and}$$

$$K_2 = F(b) - F(a) \Rightarrow \text{TPT } \frac{f'(c)}{F'(c)} = \frac{K_1}{K_2}$$

Consider a function

$$\phi(x) = K_1 F(x) - K_2 f(x) \quad \dots(1)$$

$\therefore$

$f(x)$ and $F(x)$ are continuous in $[a, b]$ and derivable in (a, b) hence $\phi(x)$ will also be continuous and differentiable

$$\text{also } \phi(a) = K_1 F(a) - K_2 f(a)$$

$$\text{and } \phi(b) = K_1 F(b) - K_2 f(b)$$

now

$$\begin{aligned} \phi(a) - \phi(b) &= K_1 (F(a) - F(b)) - K_2 (f(a) - f(b)) \\ &= [f(b) - f(a)] [F(a) - F(b)] - [F(b) - F(a)] [f(a) - f(b)] \\ &= [f(b) - f(a)] [F(a) - F(b) + F(b) - F(a)] = 0 \\ &\Rightarrow \phi(a) = \phi(b) \end{aligned}$$

$\Rightarrow$

CONTRIBUTION OF MONOTONIC BEHAVIOUR IN PLOTTING THE GRAPHS OF MISCELLANEOUS FUNCTIONS:

General tips for plotting the graph of a

rational function :

- (1) Examine whether denominator has a root or not. If no, then graph is continuous and f is non

monotonic. For eg. $f(x) = \frac{x}{x^2 - 5x + 9}$. If denominator has roots then $f(x)$ is discontinuous. Such functions can be monotonic / non-monotonic.

For eg. $f(x) = \frac{x^2 + 2x - 3}{x^2 + 2x - 8} = \frac{(x+3)(x-1)}{(x+4)(x-2)}$

- (2) If numerator and denominator has a common factor (say $x = a$) it would mean removable discontinuity at $x = a$

e.g. $f(x) = \frac{(x-2)(x-1)}{(x+3)(x-2)}$

Such a function will always be monotonic i.e. either increasing or decreasing and removable discontinuity at $x = 2$.

- (3) Compute points where the curve crosses the x -axis and also where it cuts the y -axis by putting $y = 0$ and $x = 0$ respectively and mark points accordingly.

- (4) Compute $\frac{dy}{dx}$ and find the intervals where $f(x)$ is increasing or decreasing and also where it has horizontal tangent.

- (5) In regions where curve is monotonic compute y if $x \rightarrow \infty$ or $x \rightarrow -\infty$ to find whether y is asymptotic or not.

- (6) If denominator vanishes say at $x = a$ and $(x - a)$ is not a common factor between numerator and denominator then examine

$\lim_{x \rightarrow a^+}$ and $\lim_{x \rightarrow a^-}$ to find whether

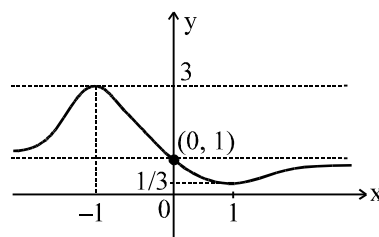
f approaches ∞ or $-\infty$. Now plot the graphs of the following functions.

ILLUSTRATIONS GRAPHS OF MISCELLANEOUS FUNCTIONS:

1. $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$; $f'(x) = \frac{2(x^2 - 1)}{(x^2 + x + 1)^2} =$

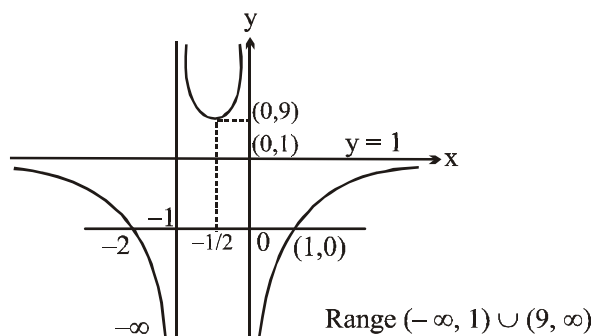
$\frac{1}{3}$; $f(-1) = 3$

$D : x \in \mathbb{R}, \text{Range} \left[\frac{1}{3}, 3 \right]$



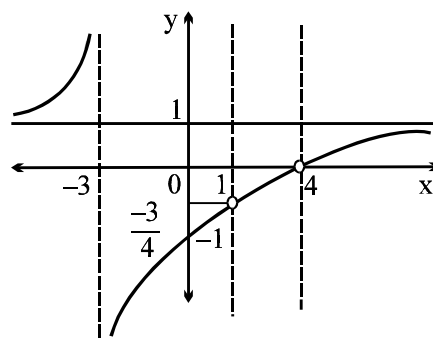
2. $f(x) = \frac{(x+2)(x-1)}{x(x+1)}$;

$f'(x) = \frac{2(2x+1)}{[x(x+1)]^2}$



3.

$f(x) = \frac{x^2 - 5x + 4}{x^2 + 2x - 3} = \frac{(x-4)(x-1)}{(x+3)(x-1)}$

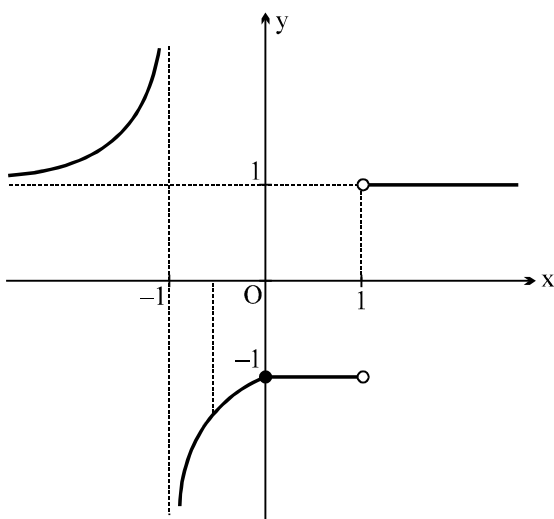


4.

$y = \frac{|x-1|}{|x|-1}$

APPLICATION OF DERIVATIVES

$$= \begin{cases} \frac{x-1}{x-1} = 1 & x > 1 \\ \text{N.D.} & \text{at } x = 1, -1 \\ \frac{1-x}{x-1} = -1 & 0 \leq x < 1 \\ \frac{1-x}{-(x+1)} = \frac{x-1}{x+1} & x < 0, x \neq -1 \end{cases}$$



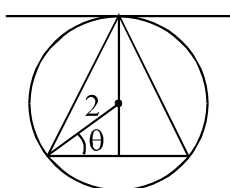
W.E-25 :- A point P is given on the circumference of a circle of radius r . Chords QR are parallel to the tangent at P. Determine the maximum possible area of the triangle PQR.

[Ans: $(3\sqrt{3}/4)r^2$] [JEE '90, 4]

Sol. Ar. of $\Delta PQR = \frac{2r \cos \theta (r + r \sin \theta)}{2}$

$$A(\theta) = r^2 \cos \theta (1 + \sin \theta)$$

$$A'(\theta) = r^2 [\cos \theta \cdot \cos \theta - (1 + \sin \theta) \sin \theta]$$



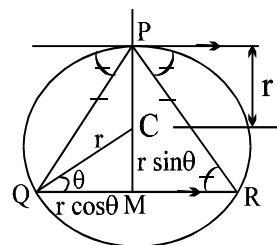
for maximum or minimum $A'(\theta) = 0$

JEE ADVANCED - VOL - III

$$\begin{aligned} \cos^2 \theta - \sin \theta (1 + \sin \theta) &= 0 \\ &= 1 - \sin^2 \theta - \sin \theta - \sin^2 \theta = 0 \\ &= 2 \sin^2 \theta + \sin \theta - 1 = 0 \\ (\sin \theta + 1)(2 \sin \theta - 1) &= 0 \\ \sin \theta &= -1 \quad (\text{not possible}) \\ \sin \theta &= 1/2 \quad \text{hence, } \theta = \pi/6 \text{ or } 5\pi/6 \end{aligned}$$

$$\frac{d^2 A}{d\theta^2} = -2 \cos \theta \sin \theta - \cos \theta - 2 \sin \theta \cos \theta$$

$$\left. \frac{d^2 A}{d\theta^2} \right|_{\theta = \pi/6} = -ve \Rightarrow A \text{ is maximum}$$



$$A_{\max} = 2r^2 \frac{\sqrt{3}}{2} \left(1 + \frac{1}{2}\right) \frac{1}{2} \Rightarrow A_{\max} = \frac{3\sqrt{3}}{4} r^2$$